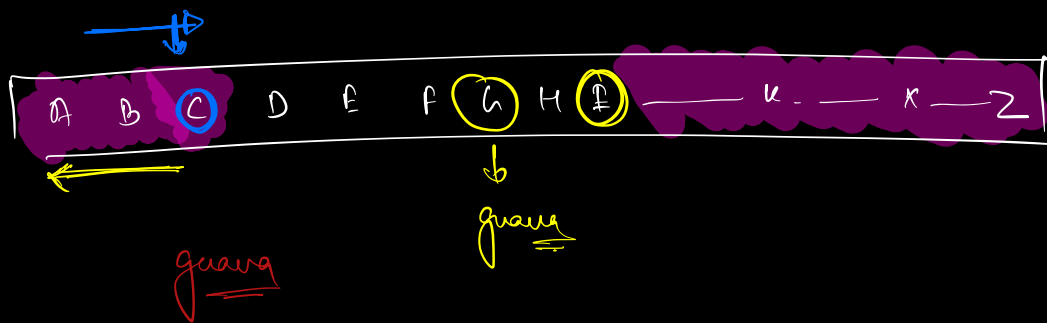


Searching :-

target → what to search? ↗
search → where to search? ↖
space

<u>target</u>	<u>search space</u>
phone no.	contacts
word	dictionary
number	array

Dictionary → 273000 words
↓



* at every step we are reducing search space.

Binary search → search element by splitting half of the search space.

Q1. Given a sorted array of size n . find if element k exists.
if it does return idx else return -1 .

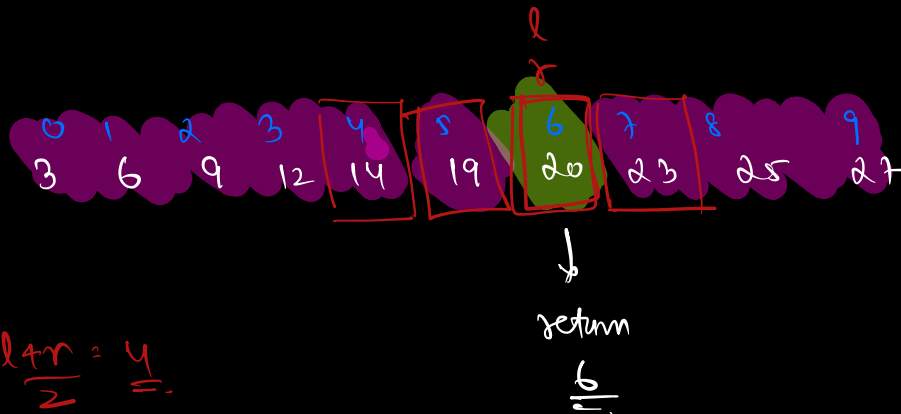
⇒

0	1	2	3	4	5	6	7	8	9
3	6	9	12	14	19	20	23	25	27

$k=20 \Rightarrow 6$

$k=21 \Rightarrow -1$

Bruteforce → Linear search
↓
 $TC \Rightarrow O(N)$



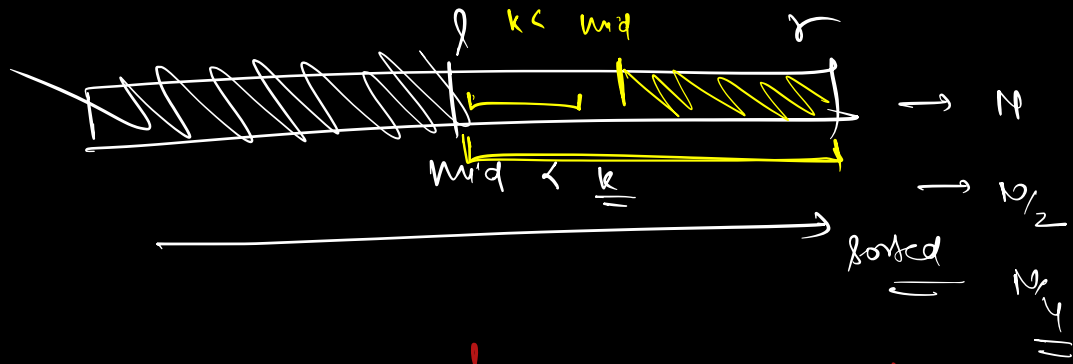
$k=20$

$$m = \frac{1+9}{2} = 5$$

$$m = \frac{5+9}{2} = 7$$

$$m = \frac{5+6}{2} = 5$$

$$m = \frac{6+6}{2} = 6$$



$k = 23$

0	1	2	3	4	5	6	7	8	9
3	6	9	12	14	19	20	23	25	27

l	r	mid	$arr[mid]$
0	9	4	$14 < 23 \rightarrow l = mid + 1$
5	9	7	$23 == 23 \rightarrow \text{return mid}$

$k = 21$

0	1	2	3	4	5	6	7	8	9
3	6	9	12	14	19	20	23	25	27

l	r	mid	$arr[mid]$
0	9	4	$14 < 21 \rightarrow l = mid + 1$
5	9	7	$23 > 21 \rightarrow r = mid - 1$
5	6	5	$19 < 21 \rightarrow l = mid + 1$
6	6	6	$20 < 21 \rightarrow l = mid + 1$

7 6

$l > r$ return -1

int search(arr[], k) {

// define search space

$l = 0, r = n-1$

while ($l \leq r$) {

int mid = $\frac{l+r}{2}$

if (arr[mid] == k) {

return mid;

} else if (arr[mid] < k) {

$l = mid + 1$

} else {

$r = mid - 1$

}

return -1;

}

TC $\Rightarrow N \rightarrow N/2 \rightarrow N/4 \rightarrow \dots \rightarrow 1$

$\underbrace{\hspace{15em}}$

$O(\log N)$

SC $\approx O(1)$

Q2. Given a sorted array, find floor of given element k .

if no floor return int_min $\rightarrow$ [greatest element $\leq k$]

arr $\Rightarrow$

0	1	2	3	4	5	6	7	8
-5	2	3	6	9	10	11	14	18

$$k = \underline{5} \rightarrow 3$$

$$k = -5 \rightarrow -5$$

$$k = 4 \rightarrow 3$$

$$k = -7 \rightarrow \underline{\text{int_min}}$$

$$k = 10 \rightarrow 10$$

$$k = 20 \rightarrow 18$$

(4) $\rightarrow$

$$\underline{\leq 11}$$

-5 2 3 6 9 10 (11)

$k \rightarrow$

greatest element $\leq k$

20 $\rightarrow$

$$\leq 20$$

[-5 2 3 6 9 10 11 14 (18)]

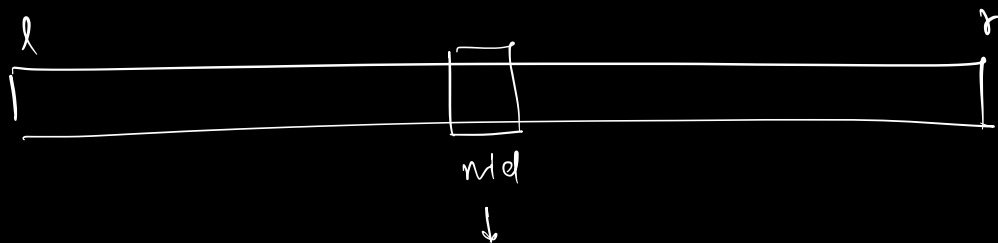
floor $\Rightarrow$ [largest] $\leq k$

$\downarrow$
18

$k = -1$ ↓
 < -1
 $(-5) \rightarrow \underline{\text{floor}}$

$k = -7$ ↓
 < -7
 $\underline{\underline{\text{return int_min}}}$

$k = -7$ ↓
 $< -7 \rightarrow \text{return int_min}$
sorted array



i) $\text{arr[mid]} == k$
return arr[mid]

ii) $\text{arr[mid]} > k$
 $r = \text{mid} - 1$

iii) $\text{arr[mid]} < k$
 $l = \text{mid} + 1$

mid
 $\downarrow$
 $[-5 \quad 2 \quad 3 \quad 6 \quad 9 \quad 10 \quad 11 \quad 14 \quad 18]$
 $\leftarrow$

floor $\Rightarrow$ [largest] $\leq k$

$k = 7$

$\Rightarrow$ Ans
 $\downarrow$
 $\text{mid} \rightarrow \text{arr}[\text{mid}] \leq k$

 $\text{arr}[\text{mid} + 1] > k$

Q3. Given a sorted array of N elements. Find the freq of elements of k .

$\text{first} \quad \text{last}$
 $\downarrow \quad \downarrow$

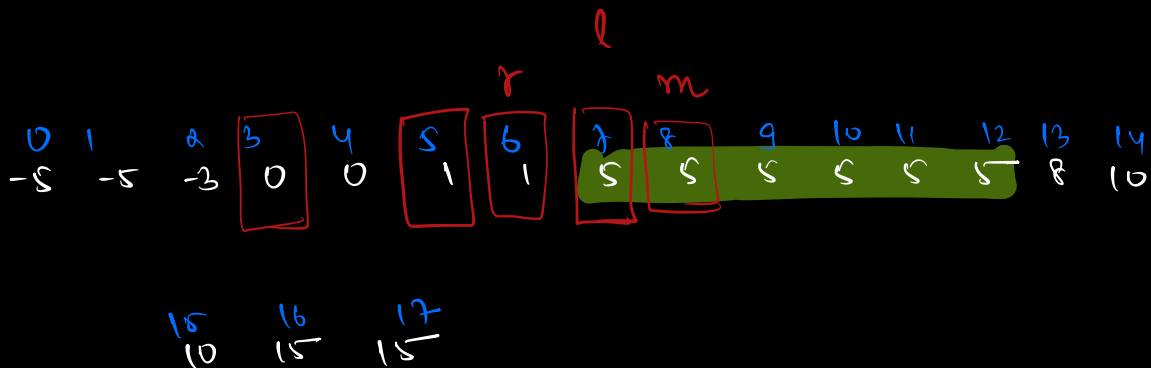
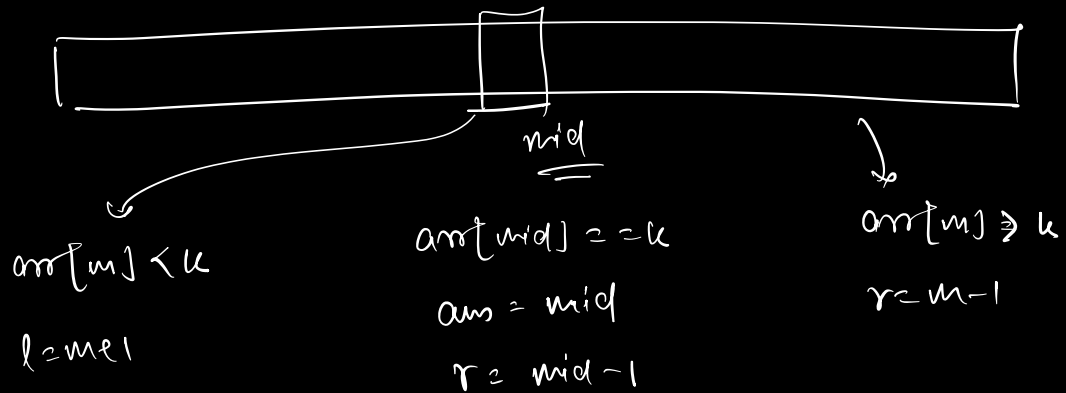
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
-5	-5	-3	0	0	1	1	5	5	5	5	5	5	8	10

15	16	17
10	15	15

$k = 5 \rightarrow \text{freq} \rightarrow \underline{\underline{6}}$

$$\text{freq} = [\text{last} - \text{first} + 1]$$

first occur



$$m = \frac{0 + 17}{2} = 8$$

$$\text{ans} = \underline{\underline{7}}$$

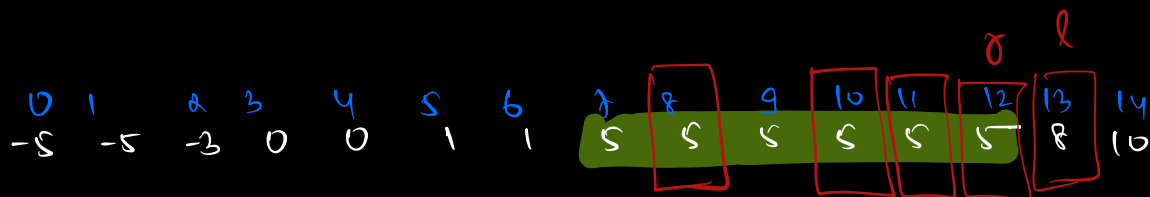
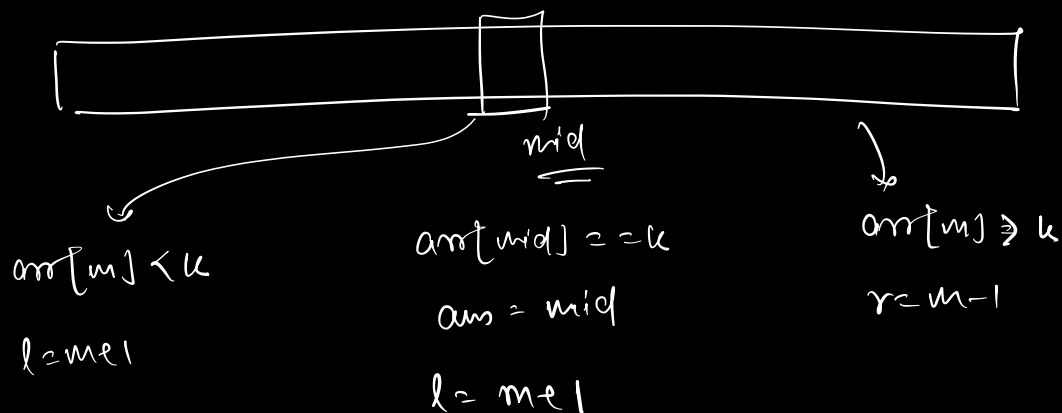
$$m = \frac{0 + 7}{2} = 3$$

$$m = \frac{4 + 7}{2} = 5$$

$$m = \frac{6 + 7}{2} = 6$$

$$m = \frac{7 + 7}{2} = 7$$

last occur



$$\begin{array}{r} 18 \\ 10 \end{array} \quad \begin{array}{r} 16 \\ 15 \end{array} \quad \begin{array}{r} 17 \\ 15 \end{array}$$

$$m = \frac{0 + 17}{2} = 8$$

$$ans = 8, 10, 11, 12$$

$$m = \frac{9 + 17}{2} = 13$$

$$m = \frac{9 + 12}{2} = 10$$

$$m = \frac{11 + 12}{2} = 11$$

$$m = \frac{12 + 12}{2} = 12$$

1st $\rightarrow$ BS

last $\rightarrow$ BS

ans $\rightarrow$ [last - first - 1]

TC $\Rightarrow O(\log N)$

SC $\Rightarrow O(1)$